§7 Collapse, blockage and collapse path sets — part 2 of 5 (external recoverability and identity, collapse as an absorbing terminal state)

Source: `paper/section7_collapse.tex`, lines 1161–2347 (`subsec:external_recovery`: the admissible external action set, preservation of identity, the external recovery predicate, relation to the state bands, to well-formed realization and to the satisfaction predicates, hand-off to the definition of collapse; `subsec:collapse_def`: the collapse indicator, relation to well-formed realization, scope of non-recoverability, consistency with external recoverability, distinction from blockage, bands and satisfaction predicates, collapse without pre-collapse bands, definition versus observability, treatment after collapse), based on commit e6e12cd, with the author-approved correction to line 2048 recorded in ledger entry L-A3.193. Status: draft. Part 1 is `07_collapse_1.md`; part 3 (`07_collapse_3.md`, 2348–3071: trajectory record space and the blockage path set) follows. §7 defines no assumption. The Japanese manuscript is authoritative; this is a summary, not a translation, and adds no definition, assumption or result.

External recoverability and identity

`Reference: subsec:external_recovery`

For states described in $\mathrm{Z}_1, \dots, \mathrm{Z}_4$, the subsection introduces auxiliary forms for describing whether *continuation as the same individual can be recovered* by an action given from outside Σ_i. This differs from autonomous response by an internal action $u \in U$. The subsection is not a theory of the physical process of external actions, of the detailed state transition after an intervention, of a general algorithm that judges identity from finite observation, or of social intervention. Its purpose is to give a descriptive boundary, before collapse is related to the concepts of §7 in `subsec:collapse_def`, between "a state recoverable as the same individual by an external action" and "a terminal state in which recovery as the same individual does not hold in the present framework".

The admissible external action set

Definition (`def:admissible_external_action`): admissible external action set (classification C). For an ACS individual Σ_i, the set of external actions considered in Phase 1 as objects of evaluation of external recoverability is written $\Upsilon_{\mathrm{adm},i}$ and called the *admissible external action set*. Each $\upsilon \in \Upsilon_{\mathrm{adm},i}$ is an action given from outside Σ_i. The set is not the set of all physically, biologically, socially or technically possible external actions; it is the set of actions included in consideration in Phase 1 to describe recoverability. Its topological, measure-theoretic and algebraic structure, duration of action, intensity, composition rules and concrete implementation are not fixed as a general form. The $\Upsilon_{\mathrm{adm},i}$ used for judging a given recoverability is fixed before that judgment and is not changed afterwards according to the course or result of the object judged. It differs in symbol, type and role from the identification state space Ξ of M1-b of §4.

Roles of u and υ are distinguished (`rem:external_action_type`). The admissible external action υ and the action $u \in U$ chosen by Σ_i itself are distinguished as variables of different roles; their non-identity is not assumed as a general condition, but υ and u are not implicitly identified. In particular υ is not implicitly substituted into the third argument or the action argument of the transition map $f(x_t, u, e_t)$. Phase 1 introduces no universal external-intervention operator saying how υ acts on f or Update. A rescue or intervention that actually took place is described on the trajectory not as an admissible external action υ but as a state change entering f through the realized external environment state e_t (background or action-response component). The treatment of re-establishment of the same individual judged on a realized trajectory follows `prop:realized_reestablishment` of §2. The admissible external action set is used only for judging recoverability.

Υ_{adm} **is a model boundary (`rem:Upsilon_adm_scope`).** $\Upsilon_{\mathrm{adm},i}$ does not mean "every means of recovery that could exist in the universe". Even if "for every $\upsilon \in \Upsilon_{\mathrm{adm},i}$, recovery does not hold" is described, physical impossibility with respect to every conceivable external action must not be derived from it. $\Upsilon_{\mathrm{adm},i}$ fixes the model-level admissible domain of actions for discussing recoverability in the Phase 1 framework. This restriction is kept when the non-recoverability of collapse is read in `subsec:collapse_def`.

Υ_{adm} **may be empty (`rem:Upsilon_adm_empty`).** This paper does not assume $\Upsilon_{\mathrm{adm},i} \neq \emptyset$ in general, so $\Upsilon_{\mathrm{adm},i} = \emptyset$ is possible depending on the system or the range of description. In that case it expresses only that "no external recovery action is specified in the present framework"; this paper does not state $\mathrm{Col}_{i,t} = 1$ from it as a general consequence. Emptiness of the admissible external action set and collapse are not identified.

Preservation of identity

Recovery versus generation of a different individual (`rem:same_identity`). *Recovery* in this subsection means only the case in which the same ACS individual Σ_i continues as the subsequent subject after the external action. If an external action forms a structure that can operate as an ACS but the subject should be treated as a distinct ACS individual $\Sigma_{i'}$ ($i' \neq i$) rather than the original Σ_i, that is not counted as recovery of Σ_i. The structural self $S_{\mathrm{struct}}(\Sigma_i)$ gives a criterion for identifying the formal structure of the same ACS individual. Phase 1 does not reduce identity to a single distance, a single continuous quantity, a single memory content or the preservation of a single physical part.

Identity judgment is not over-formalized (`rem:identity_scope`). Depending on the implementation, the structural self, the internal model, the Update structure, the self–environment boundary, continuity of memory and other things may be relevant to preserving identity. This paper does not fix all of them in Phase 1 as universal characterizing conditions of identity, and gives no universal threshold for implementation-dependent questions such as "up to what extent of part exchange is it the same individual", "what percentage of the internal model must be retained", or "how much continuity of memory is required". What Phase 1 needs is that the recovery concept distinguishes, conceptually, *renewed continuation of the same individual* from *generation of a different individual*.

The external recovery predicate

Definition (`def:external_recoverability`): external recovery predicate and external recoverability (classification C). Suppose that at time $t \in \mathbb{N}_0$ the individual Σ_i is in a state described in one of $\mathsf{Z}_1, \dots, \mathsf{Z}_4$. A deterministic binary auxiliary map $\mathrm{Recover}_{i,t} : \Upsilon_{\mathrm{adm},i} \to \{0,1\}$ is called the *external recovery predicate*. $\mathrm{Recover}_{i,t}(\upsilon) = 1$ means: under the Phase 1 recovery judgment rule fixed beforehand for the system concerned, when the external action υ is considered, the organization and coupling relations of (SC-1) of `def:collapse` are judged to be preserved or re-established as the same individual Σ_i (not as generation of a different individual), with return to subsequent operation as an ordinary ACS. $\mathrm{Recover}_{i,t}(\upsilon) = 0$ means that recoverability in this sense is not judged to hold in the present framework for that υ. The recovery judgment rule includes the criterion distinguishing the same individual from a different one and is not changed afterwards according to the course or result of the object judged; that criterion is the same one contained in the judgment rules of `def:collapse` and `def:functional_termination`. The continuation judged with υ is the continuation quantified by (SC-2) and (FT-2) with υ added, and the judgment of return to subsequent operation includes that functional activity of some basic function of (FT-1) is re-established as the same individual. The individual Σ_i is *externally recoverable* at time t if $\exists \upsilon \in \Upsilon_{\mathrm{adm},i} : \mathrm{Recover}_{i,t}(\upsilon) = 1$.

The predicate is not the dynamics of intervention (`rem:recover_not_intervention_dynamics`). $\mathrm{Recover}_{i,t}$ is not a state-transition map that explicitly generates the physical process after the external action. Phase 1 introduces no universal external-intervention transition map such as $(x_t, \upsilon) \mapsto x'$. It is a classification-C auxiliary predicate for describing external recoverability determined for each system. The detailed state trajectory after intervention, intensity, composition of several actions, cost and reachability of the intervention are not included in the common form of this subsection.

Recover = 1 **does not mean that recovery actually occurs (`rem:recoverability_not_actual_recovery`).** $\mathrm{Recover}_{i,t}(\upsilon) = 1$ says that recovery as the same individual is *judged possible* under υ. It does not mean that υ is actually given after t or that recovery occurs on an actual trajectory. Possibility of recovery, carrying out of a recovery action and actual attainment of recovery are distinguished as possibility, execution and result; external recoverability is not identified with a future realized event.

Interpretation of Recover = 0 **(`rem:recover_zero_scope`).** $\mathrm{Recover}_{i,t}(\upsilon) = 0$ says that recovery by υ is not recognized to hold under the judgment rule of the present framework; it does not mean that recovery by υ is impossible in a cosmic or physical sense. Even if $\mathrm{Recover}_{i,t}(\upsilon) = 0$ for all $\upsilon \in \Upsilon_{\mathrm{adm},i}$, collapse is not defined from that alone: collapse requires (SC-1) and (SC-2) of `def:collapse`.

Relation to the descriptive pre-collapse state bands

Scope of the predicate (`rem:recover_predicate_scope`). External recoverability is used for states in the descriptive pre-collapse bands $\mathrm{Z}_1, \dots, \mathrm{Z}_4$. The normal-operation reference band Z_0 is not treated as an object of "recovery". That maintaining Z_0 may require external resources or support is a separate problem from recoverability from a disordered state. For the collapse band Z_5, the recoverability definition here is not applied to deny collapse; it is left to the definition of collapse in §2 (`def:collapse`).

Band numbers and recoverability (`rem:zone_not_recoverability_order`). This paper does not state, from the size of the indices of $\mathrm{Z}_1, \dots, \mathrm{Z}_4$, any ordering of recoverability or of probability of recovery as a general consequence. The numbering convention does not exclude, for example, that a blockage state Z_4 is recoverable by some admissible external action while some partial-malfunction state Z_2 is not judged recoverable in the present framework. The Z bands are not a scale of difficulty of recovery.

Blockage and external recoverability (`rem:blockage_external_recovery`). Blockage $A_{i,t} = \emptyset$ is the time-local one-step condition corresponding to Z_4. Its holding does not deny external recoverability: even with $A_{i,t} = \emptyset$ at time t, the case $\exists \upsilon \in \Upsilon_{\mathrm{adm},i} : \mathrm{Recover}_{i,t}(\upsilon) = 1$ is allowed. This gives the recovery-side description of the treatment in `subsec:blockage` that blockage does not give $\mathrm{Col}_{i,t} = 1$ as a general consequence.

Relation to well-formed realization and the satisfaction predicates

External recoverability and well-formed realization (`rem:recovery_and_wf`). External recoverability is not synonymous with $\mathrm{WF}_{i,t} = 0$. Even when Σ_i in a pre-collapse band keeps well-formed realization at time t, functional recovery by an external action can be meaningful. Conversely this paper does not state the presence or absence of external recoverability from $\mathrm{WF}_{i,t} = 0$ alone as a general consequence. $\mathrm{WF}_{i,t}$ and $\mathrm{Recover}_{i,t}$ are different formal objects; the definition of collapse (`def:collapse`) also distinguishes non-realization of well-formed realization from non-recoverability relative to the present framework.

External recoverability and the satisfaction predicates (`rem:recovery_not_principle_truth`). External recoverability is not a condition of any time-indexed satisfaction predicate of SM, EU or FR. From $\exists \upsilon : \mathrm{Recover}_{i,t}(\upsilon) = 1$, it does not follow that the SM, EU or FR predicates are true at t. Conversely, even if the three predicates are true at t, the state may not require an external recovery action, or $\Upsilon_{\mathrm{adm},i}$ may be empty. External recoverability is not the judgment of the three satisfaction predicates.

Hand-off to the definition of collapse

Direction of dependence (`rem:recovery_collapse_distinction`). The subsection defines external recoverability first and does not define collapse itself. The logical direction of dependence is: admissible external action domain, then external recoverability, then reference to non-recoverability in the definition of collapse. Condition (SC-2) of `def:collapse` refers to $\Upsilon_{\mathrm{adm},i}$ of this subsection, and `subsec:collapse_def` reads that non-recoverability together with the scope of external recoverability. The definition of $\mathrm{Recover}_{i,t}$ does not use the truth value of $\mathrm{Col}_{i,t}$, which avoids the circular definition "recoverability ↔ not being collapse".

Recoverable implies not collapse (`rem:recoverable_excludes_collapse`). For a time t in $Z_1, \ldots, Z_4$, if $\exists \upsilon \in \Upsilon_{\mathrm{adm},i} : \mathrm{Recover}_{i,t}(\upsilon) = 1$, the state at that time is not a collapse state (`prop:recover_excludes_termination`). This does not derive a new definition of collapse from external recoverability; it follows from the requirement of consistency with (SC-2) imposed on the recovery judgment rule (`def:external_recoverability`). The distinction "recoverable as the same individual in the present framework implies not treated as collapse" is maintained.

Phase 1 limits of external recoverability (`rem:external_recovery_phase1_scope`). Five limits are stated. First, $\Upsilon_{\mathrm{adm},i}$ does not cover all external actions. Second, $\mathrm{Recover}_{i,t}$ is a classification-C auxiliary predicate abstracting a system-specific judgment rule and not a map generating post-intervention transitions. Third, Phase 1 gives no universal characterizing conditions of identity. Fourth, existence of recoverability does not mean execution of a recovery action or actual attainment of recovery. Fifth, this paper does not state collapse from Recover = 0 or $\Upsilon_{\mathrm{adm}} = \emptyset$ alone as a general consequence. The subsection is therefore not a theory that decides collapse and non-collapse from finite observation; it gives the *conceptual boundary on the recoverability side* referred to by (SC-2) of the definition of collapse.

Collapse: an absorbing terminal state in the present framework

`Reference: subsec:collapse_def`

The subsection relates the functional termination and collapse of Σ_i defined in §2 (`def:functional_termination`, `def:collapse`) to the concepts of §7.

The collapse indicator

The functional-termination judgment time domain $\mathbb{T}_{\mathrm{FT},i}$, the functional-termination state indicator $\mathrm{FTS}_{i,t}$ with its conditions (FT-1), (FT-2), the functional-termination indicator $\mathrm{FT}_{i,t}$, the collapse-state indicator $\mathrm{CS}_{i,t}$ with its conditions (SC-1), (SC-2), and the collapse indicator $\mathrm{Col}_{i,t} \in \{0, 1\}$ are given in §2. Absorption (C-2) and existence of a time at which the collapse state holds (C-3) are in `prop:collapse_basic` of §2; `prop:termination_order` establishes that $\mathrm{Col}_{i,t} = 1$ implies $\mathrm{FT}_{i,t} = 1$: a functional-termination state has occurred by time t (the history indicator; the state indicator $\mathrm{FTS}_{i,t}$ is not claimed to be 1 at t). Non-recoverability of collapse means (SC-2), and that of functional termination means (FT-2). The subsection does not redefine them and reads the content of $\Upsilon_{\mathrm{adm},i}$, to which (FT-2) and (SC-2) refer, according to `def:admissible_external_action`.

Meaning of the value 0 (`rem:collapse_zero_meaning`). $\mathrm{Col}_{i,t} = 0$ says that at no time up to t has the collapse state in the sense of `def:collapse` held. This paper does not state from it, as a general consequence, $\mathrm{WF}_{i,t} = 1$, $A_{i,t} \neq \emptyset$, $x_t \in K$ or truth of the SM, EU or FR predicates. $\mathrm{Col}_{i,t} = 0$ is not used as a general indicator of normality such as "healthy", "feasible" or "recovered".

Relation to well-formed realization

`rem:wf_failure_not_collapse`. The relation of the functional-termination and collapse indicators to the well-formed realization indicator follows `rem:collapse_wf_relation` of §2.

Collapse and the structural self (`rem:collapse_structural_self`). The structural self $S_{\mathrm{struct}}(\Sigma_i)$ is a criterion identifying the formal structure of the same ACS individual. $\mathrm{WF}_{i,t} = 0$ at and after a time at which collapse holds (`prop:termination_order`) expresses that the structural self is not well-formedly realized at t. The converse, collapse from $\mathrm{WF}_{i,t} = 0$, is not stated as a general consequence (`rem:collapse_wf_relation`). The formal descriptive object "structural self" is not thereby eliminated as a mathematical object. Collapse does not mean that "the description of the 7-tuple ceases to exist in the paper", but that ordinary realization and continuation as that same ACS individual have been made terminal within the present framework.

Scope of non-recoverability

Non-recoverability of collapse is relative to the present framework (`rem:collapse_relative_irrecoverability`). The non-recoverability of (SC-2), and that of (FT-2), is relative to the admissible external action domain fixed by $\Upsilon_{\mathrm{adm},i}$ and to the range of description of continuation by ordinary autonomous update and by g_O under the realized sequence of exogenous background components (`def:collapse`, (SC-2)). Hence this paper does not claim that, from $\mathrm{Col}_{i,t} = 1$, recovery as the same individual is impossible without qualification even by all conceivable physical actions, unknown technology, operations outside Phase 1, or interventions not modeled. What it claims is the limited terminality "non-recoverable within the range of admissible continuations of the present framework".

$\Upsilon_{\mathrm{adm}} = \emptyset$ and collapse (`rem:empty_Xi_not_collapse`). This paper allows $\Upsilon_{\mathrm{adm},i} = \emptyset$ but does not state $\mathrm{Col}_{i,t} = 1$ from the fact that no admissible external action is specified. Collapse requires at least that (SC-1) loss of the organizational relations that establish structural identity and (SC-2) non-recoverability as the same individual relative to the present framework hold together at some time. Emptiness of the external action domain and collapse are different conditions.

The universal statement Recover = 0 **and collapse (`rem:all_recover_zero_not_collapse`).** Even if in a pre-collapse band $\mathrm{Recover}_{i,t}(\upsilon) = 0$ held for all $\upsilon \in \Upsilon_{\mathrm{adm},i}$, this paper does not state collapse from that fact alone. First, $\mathrm{Recover}_{i,t}$ is a classification-C auxiliary predicate on the recoverability side, not an indicator judging collapse. Second, collapse requires the loss of organizational relations of (SC-1). Third, (SC-2) is a terminality condition on the whole of admissible continuations of the same individual and is not reduced to the value of $\mathrm{Recover}_{i,t}$ at a single time. Thus "external recoverability not confirmed" is not "collapse".

Consistency with external recoverability

Proposition (`prop:recover_excludes_termination`): external recoverability and the terminal state indicators. If $\mathrm{Recover}_{i,t}$ is defined at time t and $\mathrm{Recover}_{i,t}(\upsilon) = 1$ for some $\upsilon \in \Upsilon_{\mathrm{adm},i}$, then $\mathrm{CS}_{i,t} = 0$. Moreover, if $t \in \mathbb{T}_{\mathrm{FT},i}$, then $\mathrm{FTS}_{i,t} = 0$.

Proof (outline). By `def:external_recoverability`, on one continuation with υ added, and under the same identity criterion as (SC-2) and (FT-2), the organization and coupling relations of (SC-1) are preserved or re-established, and functional activity of some basic function of (FT-1) is re-established as the same individual. If preserved, (SC-1) does not hold at t; if re-established, re-establishment is recognized on a continuation quantified by (SC-2), so (SC-2) does not hold. In either case $\mathrm{CS}_{i,t} = 0$. Likewise that continuation is one quantified by (FT-2) and basic-function activity is re-established on it, so for $t \in \mathbb{T}_{\mathrm{FT},i}$ (FT-2) does not hold and $\mathrm{FTS}_{i,t} = 0$. □

Recoverable states are not treated as collapse (`rem:C2_and_recoverability`). For a time t in $\mathrm{Z}_1, \ldots, \mathrm{Z}_4$ with $\exists \upsilon \in \Upsilon_{\mathrm{adm},i} : \mathrm{Recover}_{i,t}(\upsilon) = 1$, the state is not treated as a collapse state at t. By `prop:recover_excludes_termination` such a state is not a collapse state and, if the time belongs to the functional-termination judgment time domain, not a functional-termination state either. This is consistent with the dependence direction fixed in `subsec:external_recovery`, and since $\mathrm{Col}_{i,t}$ is not used in defining external recoverability, no circular definition arises.

Distinction from blockage, bands and the satisfaction predicates

Limits of the implication between blockage and collapse (`rem:collapse_non_implication`). Blockage is the non-existence of an action retaining one-step feasibility under e_t; it differs from the functional-termination conditions (FT-1), (FT-2) and from the collapse-state conditions (SC-1), (SC-2). This paper does not state collapse from blockage as a general consequence (`rem:blockage_not_collapse_sec2` of §2). In particular, in the blockage state Z_4, if external recoverability exists the state is not collapse. Nor is prior holding of blockage imposed as a requirement for collapse.

Total stoppage and collapse (`rem:full_stop_not_collapse`). The total-stoppage band Z_3 is an auxiliary band describing stoppage of the operation of the standard functions relating to TIPA, PSL and ISL. From total stoppage alone it does not follow that (FT-1) and (FT-2) both hold, nor that (SC-1) and (SC-2) both hold. This paper does not state $\mathrm{FT}_{i,t} = 1$ from being described as Z_3 alone as a general consequence. Z_3 is a description of the operation of the

standard functional forms (classification B), whereas (FT-1) is a condition on stoppage of the basic and base functions held by the individual for each division of primitive functions; the two differ in level of functional description, and the section places no implication in either direction. Hence it does not state $\mathrm{Col}_{i,t} = 1$ from Z_3 alone; in particular it does not exclude that a totally stopped individual is recoverable as the same individual.

A false satisfaction predicate and collapse (`rem:principle_failure_not_collapse`). This paper does not state $\mathrm{Col}_{i,t} = 1$ as a general consequence from one or several of the SM, EU and FR predicates being false at t. The three principles are constraint axes defined in §3 and their predicates are conditions on the state and environment of a persisting ACS, whereas collapse is the terminal of the structural individual defined in §2. The truth values of the predicates are therefore distinct from the collapse judgment. Nor does it state collapse from failure of T-STAB, relation quality $Q < 0$, or $\chi_{\mathrm{EI},t} = 0$ alone.

Collapse without passing through the pre-collapse bands

Prior passage is not a requirement (`rem:collapse_without_precollapse`). `def:collapse` does not require, before collapse $\mathrm{Col}_{i,t} = 1$ holds, that any of $\mathrm{Z}_1, \mathrm{Z}_2, \mathrm{Z}_3, \mathrm{Z}_4$ has been observed or described. Trajectories that reach the collapse terminal in the sense of §7 without explicitly passing through blockage, partial malfunction or total stoppage are not excluded from the general Phase 1 theory. Conversely it is not claimed that a trajectory in $\mathrm{Z}_1, \dots, \mathrm{Z}_4$ reaches collapse in every case. The distinction is maintained in `subsec:col_path_set` and `subsec:inflow_paths`.

Definition of collapse versus observability

They differ (`rem:collapse_definition_observability`). The absorption (C-2) of `prop:collapse_basic` and the non-recoverability (SC-2) of `def:collapse` concern future admissible continuations, not the present time alone. Phase 1 therefore gives no universal procedure that judges $\mathrm{Col}_{i,t} = 1$ in general and deterministically from observation at a finite time. This does not mean that the formal definition of collapse is unclear: the paper distinguishes the definition of the terminal state in theory from a judgment method recognizing it from finite observation. Implementation-dependent sufficient conditions, diagnostic indicators, observation windows, misjudgment rates and probabilistic estimation required for the latter are tasks for Phase 2 and later (L9 of `subsec:limitations_list`, `prob:collapse_observability`).

The indicator need not be an internal variable (`rem:collapse_indicator_description_level`). $\mathrm{Col}_{i,t}$ is a theoretical indicator introduced to describe the collapse state. Σ_i is not required to hold, compute or recognize that same binary variable internally, and M2 or other mechanisms reading the true $\mathrm{Col}_{i,t}$ directly is not a Phase 1 requirement. The indicator is a formal representation on the description and diagnosis side and is not put in one-to-one correspondence with a specific implementation variable.

Ordinary ACS state quantities after collapse

No ordinary ACS dynamics is required after collapse (`rem:collapse_internal_state`). After $\mathrm{Col}_{i,t} = 1$ holds, the collapse indicator is retained as a terminal record by (C-2). For times after collapse, describing Σ_i as an ordinary ACS by $x_s \in X$, $e_s \in E_{\mathrm{ext}}$, $A_{i,s}$ and Desc_s is in general not required. This does not mean that the physical object literally disappears. It means that the Phase 1 ACS trajectory description does not bear the duty to describe the subsequent period as the ordinary internal dynamics of the same ACS individual. This terminal convention is formally reflected in the trajectory record space introduced in `subsec:blk_path_set`.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.